

YUNHAI XIANG

Mathematician · Data Scientist · Software Engineer

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EDUCATION

Western University (University of Western Ontario)

2024 — Now

Ph.D. (in progress) and M.Sc. in Mathematics

London, ON, Canada

Advisor: Chris Hall

Specialization: Number Theory and Arithmetic Geometry

University of Waterloo

2018 — 2023

B.Math. in Pure Mathematics, Honors, with Distinction

Waterloo, ON, Canada

INDUSTRY EXPERIENCE

ByteDance

2019

Co-op Internship

Beijing, China

- Supported data-informed product development for an education technology product
- Collected and reviewed educational data to support product analysis and application improvement

SKILLS

Colored by level of mastery: *Expert, Advanced, Proficient, Familiar, Beginner*

Languages

[Mandarin](#), [English](#), French, Japanese

Programming

[Python](#), [Lean](#), [C/C++](#), C#, Rust, Haskell, JavaScript/TypeScript

Computing

[SageMath](#), [NumPy](#), [SciPy](#), [Pandas](#), [PyTorch](#), [XGBoost](#), [Sklearn](#), [Slurm](#)

DSLs

[LaTeX](#), [SQL](#), [HTML](#), [CSS](#)

Softwares

[Cinema 4D](#)

Hobbies

[Magic Tricks](#)

PUBLICATIONS

QEDBench: Quantifying the Alignment Gap in Automated Evaluation of University-Level Mathematical Proofs

Joint with Santiago Gonzalez, Alireza Amiri Bavandpour, Peter Ye, Edward Zhang, et al.

INTERNATIONAL CONFERENCE ON MACHINE LEARNING (ICML) 2026

AWARDS

Mathematics Undergraduate Research Award

2022

- Highly competitive undergrad research award of value \$10000 at University of Waterloo. Worked on research project in summer related to Weyl groups, supervised by Prof. Matthew Satriano.

ACADEMIC MEETINGS & PRESENTATIONS

Conferences Attendance & Presentations

Attended academic conferences and delivered research or expository talks

- Workshop on Noncommutative Geometry and Graph Algebras, Summer 2026, Western
- FrontierMath Open Problems Workshop, Summer 2026, UToronto

- Homotopy Theory, K-theory, and Topological Data Analysis Conference, Summer 2026, Western
- Using AI to Find Examples Workshop, under thematic program Universal Statistics in Number Theory, Winter 2026, Centre de Recherches Mathématiques, Université de Montréal
- University of Western Ontario Research in Computer Science (UWORCS), Winter 2026, Western
- *Rigid Analytic Geometry*, Ontario Graduate Mathematics Conference, Summer 2024, UWaterloo
- *Cohomology of Grassmannians*, Canadian Undergraduate Mathematics Conference, Summer 2022, Université Laval

Seminar Attendance & Presentations

Attended academic seminars and delivered research or expository talks

- *Hyperelliptic Jacobians and Hasse–Witt Matrix*, Algebraic Geometry Student Seminar, Western
- *Trace of Frobenius*, Winter 2026, Graduate Student Seminar, Western
- *Brauer Groups of Fields Part II*, Winter 2026, Algebraic Geometry Student Seminar, Western
- *Brauer Groups of Fields Part I*, Winter 2026, Algebraic Geometry Student Seminar, Western
- *Weyl Group and Root Datum*, Summer 2025, Algebraic Groups Seminar, Western
- *Hecke Operators*, Summer 2025, Algebraic Groups Seminar, Western
- *Parabolic and Borel Subgroups*, Summer 2025, Algebraic Groups Seminar, Western
- *Lie Algebras of Algebraic Groups*, Summer 2025, Algebraic Groups Seminar, Western
- *Admissible Topology*, Summer 2024, Rigid Analytic Geometry Seminar, Western
- *Non-Archimedean Fields*, Summer 2024, Rigid Analytic Geometry Seminar, Western
- *Cones and Affine Toric Varieties*, Summer 2024, Toric Varieties Learning Seminar, Western
- *An Easy Tour of Galois Cohomology*, Winter 2024, Graduate Student Seminar, Western
- *Introduction to Derived Categories*, Fall 2022, Algebraic Geometry Working Seminar, UWaterloo

TEACHING & MENTORING

Course Instructorship

- Coming soon.

Teaching Assistantship

Proctored and graded exams, graded homeworks, tutored at Help Centre, and taught tutorial sessions

- MATH 3150, *Elementary Number Theory I*, Fall 2024
- MATH 2155, *Mathematical Structures*, Fall 2025
- CALCULUS 1501, *Calculus II for Mathematical and Physical Sciences*, Winter 2025, Winter 2026, Summer 2026
- MATH 1229, *Methods of Matrix Algebra*, Summer 2024, Summer 2025
- MATH 1228, *Methods of Finite Mathematics*, Summer 2024, Summer 2025
- CALCULUS 1000, *Calculus I*, Winter 2024, Summer 2024, Winter 2025, Fall 2025, Winter 2026

Mathematical Outreach Lectures

Gave expository lectures to high school students or undergraduate students on advanced mathematical topics

- *Aperiodic Tilings*, Summer 2026, Western Summer Math Camp (Grade 11/12)
- *Surreal Numbers and the Game of Go*, Summer 2025, Western Summer Math Camp (Grade 10)
- *Mathematical Magic Tricks: Gilbreath's Principle*, Summer 2024, Western Summer Math Camp (Grade 11)
- *Why is There no Quintic Formula*, Summer 2024, Western Summer Math Camp (Grade 10)
- *Category Theory Demystified*, Winter 2023, Short Attention Span Math Seminar (Undergraduate)

Directed Reading Program Mentorship

Mentored undergraduate students on reading projects, culminating in a presentation

- Jude McNorgan Lubos, *Introduction to Quadratic Reciprocity*, Winter 2026

SERVICE & VOLUNTEER WORK

Seminar Organization

Organized or co-organized research or learning seminars

- Algebraic Geometry Student Seminar, Summer 2026
- Algebraic Geometry Student Seminar, Winter 2026, co-organized with Alex Zwart
- Algebraic Geometry Student Seminar, Fall 2024, co-organized with Alex Zwart
- Toric Varieties Learning Seminar, Summer 2024, co-organized with Nikolas Rieke

SOFTWARE PROJECTS & OPEN SOURCE CONTRIBUTIONS

Coming soon.

CERTIFICATIONS

Coming soon.